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Lab Report

Course Title: Internet of Things Lab

Course Code: CSE 402

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Experiment no: 05

Experiment Name: Interfacing of temperature sensor LM35 with Arduino.

Introduction

The LM35 series are precision integrated-circuit temperature devices with output voltage linearly proportional to the Centigrade temperature. LM35 is three terminal linear temperature sensors from National semiconductors. It can measure temperature from -55 degree Celsius to +150 degree Celsius. The voltage output of the LM35 increases 10mV per degree Celsius rise in temperature. LM35 can be operated from a 5V supply and the stand by current is less than 60uA. The pin out of LM35 is shown in the figure below.

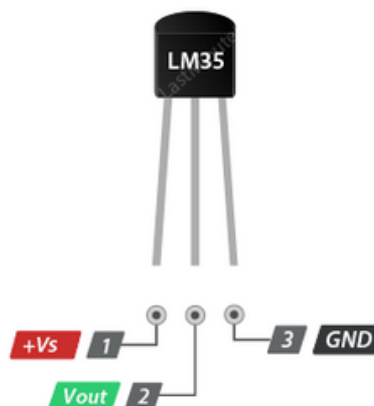
LM35 Sensor Pinout

The LM35 comes in three different form factors, but the most common type is the 3-pin TO-92 package, which looks just like a transistor. Let's take a look at its pinout.

+Vs is the power supply for the sensor which can be anywhere between 4V to 30V.

Vout pin produces an analog voltage that is directly proportional (linear) to the temperature. It should be connected to an Analog (ADC) input.

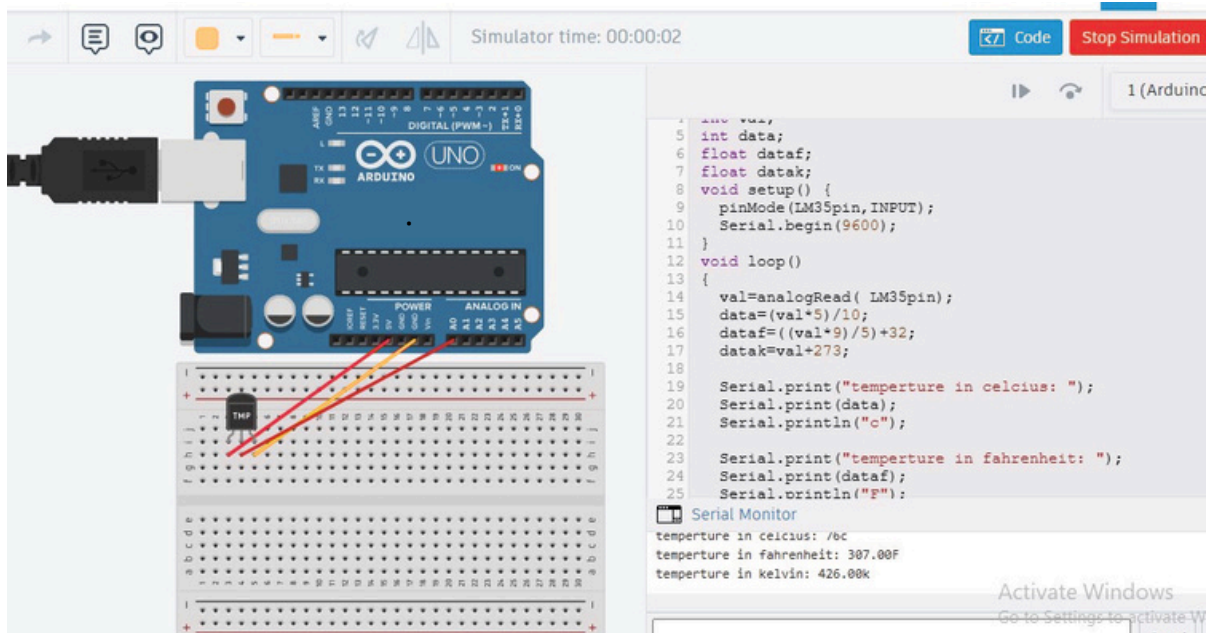
GND is a ground pin.



Hardware Required:

Component Name	Quantity
Arduino UNO	1
Lm35	1
USB Cable	1
Breadboard	1
Jumper wires	Several

Circuit Diagram:



Steps of working

1. Insert the temperature sensor into your breadboard and connect its pin! the supply.
2. Connect its center pin to the analog pin A) and the remaining pin3 to GND on the breadboard.
3. Upload the code as given below.
4. Vary the temperature and read the voltage changes.
5. Open the Arduino IDE's serial monitor to see the results

Connecting the LM35 Temperature Sensor to an Arduino :

Hooking up the LM35 to an Arduino is super simple. You only need to connect three pins: two for power and one for reading the sensor value.

The sensor can be powered from 5V. The positive voltage connects to '+Vs' and ground connects to 'GND'. The middle pin 'Vout' is the analog signal output from the sensor and connects to the A0 analog input of an Arduino.

The Sketch

This sketch works by setting pin A0 as for the temperature semor. After that the run a loop that continually reads the value from the sensor and sends that value voltage. The voltage value is between 0-5 volts, when temperature will vary accordingly.

Code:

```
const int LM35pin = A0;
float temperature;
int val;
int data;
float dataf;
float datak;

void setup() {
  pinMode(LM35pin,INPUT);
  Serial.begin(9600);
}
void loop()
{
  val=analogRead( LM35pin);
  data=(val*5)/10;
  dataf=((val*9)/5)+32;
  datak=val+273;

  Serial.print("temperture in celcius: ");
  Serial.print(data);
  Serial.println("c");

  Serial.print("temperture in fahrenheit: ");
  Serial.print(dataf);
  Serial.println("F");

  Serial.print("temperture in kelvin: ");
  Serial.print(datak);
  Serial.println("k");

  delay(3000);
}
```

Observation Table:

sr.no:	voltage	temperature
1	25	celcius: 76c fahrenheit: 307.00F kelvin: 426.00k
2	80	celcius: 76c fahrenheit: 307.00F kelvin: 426.00k

Conclusion:

This experiment successfully demonstrated the LM35 temperature sensor's functionality with an Arduino, providing accurate temperature readings. The Arduino's ADC effectively converted the sensor's analog output, proving the setup's viability for basic temperature monitoring.